



Solution Set

Question 1

A Black-Scholes implied volatility

$$\sigma$$

solves

$$C_{BS}(\sigma) = C_{mkt}$$

Write Newton's method for sigma and explain why vega appears in the denominator.

Answer key

Define

$$f(\sigma) = C_{BS}(\sigma) - C_{mkt}$$

Newton's method is

$$\sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)}$$

Since

$$f'(\sigma) = \frac{\partial C_{BS}}{\partial \sigma} = \nu$$

the update is

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{\nu(\sigma_n)}$$

Vega converts a price error into an implied-volatility correction.

Question 2

Do two Newton iterations to solve

$$x^2 - 2 = 0$$

from

$$x_0 = 1$$

Use this as an analogy for implied-volatility inversion.

Answer key

For $f(x) = x^2 - 2$, $f'(x) = 2x$. The update is

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$$

$$x_1 = \frac{1}{2} (1 + 2) = 1.5$$

$$x_2 = \frac{1}{2} \left(1.5 + \frac{2}{1.5} \right) = 1.4167$$

Like implied-volatility solving, a nonlinear equation is converted into repeated local linear corrections.

Question 3

A least-squares calibration minimizes

$$F(\theta) = \frac{1}{2} \|r(\theta)\|^2$$

Derive the Gauss-Newton step.

Answer key

Linearize residuals around theta:

$$r(\theta + \Delta) \approx r(\theta) + J\Delta$$

Minimize the quadratic approximation

$$\frac{1}{2} \|r + J\Delta\|^2$$

The normal equations are

$$J^\top J\Delta = -J^\top r$$

so the Gauss-Newton step solves

$$\Delta = -(J^\top J)^{-1} J^\top r$$

provided $J^\top J$ is invertible or suitably regularized.

Question 4

Explain why Levenberg-Marquardt damping is useful in volatility-surface calibration.

Answer key

Damping replaces $J^\top J$ by $J^\top J + \lambda I$. The step becomes

$$(J^\top J + \lambda I)\Delta = -J^\top r$$

When lambda is large, the step behaves more like stable gradient descent; when lambda is small, it behaves like Gauss-Newton. This helps when parameters are highly correlated or the local quadratic approximation is unreliable.

Question 5

For

$$f(x, y) = 100(y - x^2)^2 + (1 - x)^2$$

why is vanilla gradient descent difficult?

Answer key

This Rosenbrock-style objective has a curved narrow valley. Gradients can point strongly across the valley instead of along it, so a large learning rate overshoots and a small learning rate crawls. The interview point is conditioning: curvature differs dramatically across directions.

Question 6

Compute one gradient-descent step for $f(x,y)=(x-3)^2+10(y+1)^2$ from (0,0) with learning rate 0.05.

Answer key

The gradient is

$$\nabla f = (2(x-3), 20(y+1))$$

At (0,0),

$$\nabla f = (-6, 20)$$

The update is

$$(x_1, y_1) = (0, 0) - 0.05(-6, 20) = (0.3, -1.0)$$

Notice the y coordinate jumps exactly to its optimum because curvature in y is high and the step size happens to match it.

Question 7

Why can a calibration objective be small while the model is still bad for market risk?

Answer key

A small average squared pricing error can hide concentrated errors in tail strikes, long maturities, or illiquid but risk-important instruments. Market risk cares about sensitivities and stress behavior, not just average fit. A serious answer checks weighted errors, Greeks, arbitrage constraints, and out-of-sample stability.

Question 8

Derive the normal equations for minimizing

$$\|X\beta - y\|^2$$

Answer key

Let

$$F(\beta) = (X\beta - y)^\top (X\beta - y)$$

Differentiate:

$$\nabla_\beta F = 2X^\top (X\beta - y)$$

Set the gradient to zero:

$$X^\top X\hat{\beta} = X^\top y$$

If $X^\top X$ is invertible,

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

Question 9

Explain line search in the context of Newton or quasi-Newton calibration.

Answer key

A method may propose a direction p_k . Line search chooses a step length α_k so the objective decreases sufficiently:

$$\theta_{k+1} = \theta_k + \alpha_k p_k$$

This prevents a good local direction from taking an excessively large step that worsens calibration or violates bounds.

Question 10

What is the difference between parameter bounds and parameter transformations? Give an example.

Answer key

Bounds constrain the optimizer directly, such as $\sigma \geq 0$. A transformation reparameterizes the variable so the constraint is automatic:

$$\sigma = e^\theta$$

The transformed optimizer is unconstrained in θ but always produces positive volatility. Transformations can improve feasibility but may change scaling and conditioning.

Question 11

A Hessian has eigenvalues 0.001 and 100. What does this imply for numerical optimization?

Answer key

The condition number is $100 / 0.001 = 100000$. The objective is extremely flat in one direction and very steep in another. Gradient methods may zigzag, Newton steps may be sensitive to noise, and regularization or rescaling may be needed.

Question 12

Describe a robust calibration validation checklist after an optimizer converges.

Answer key

Check convergence criteria, residual plots, parameter bounds, arbitrage constraints, sensitivity stability, finite-difference agreement, out-of-sample fit, multiple starting points, and economic plausibility. In interviews, convergence alone is never the end of the calibration story.